CSI BISHOP SOLOMON DORAISAWMY COLLEGE OF ARTS AND SCIENCE

Department of Computer Science

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Project Development Phase

Step-by-Step Implementation Guide

Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

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Platform	ServiceNow

1. Development Overview

This document details the complete step-by-step implementation of the project, including data preparation, import set configuration, transform map creation, and dot walking implementation.

Project Title	Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking
Document Type	Project Development Document
Platform	ServiceNow PDI
Total Milestones	2 (Data Import + Dot Walking)
Total Activities	6

Milestone 1: Import Data from Data Source

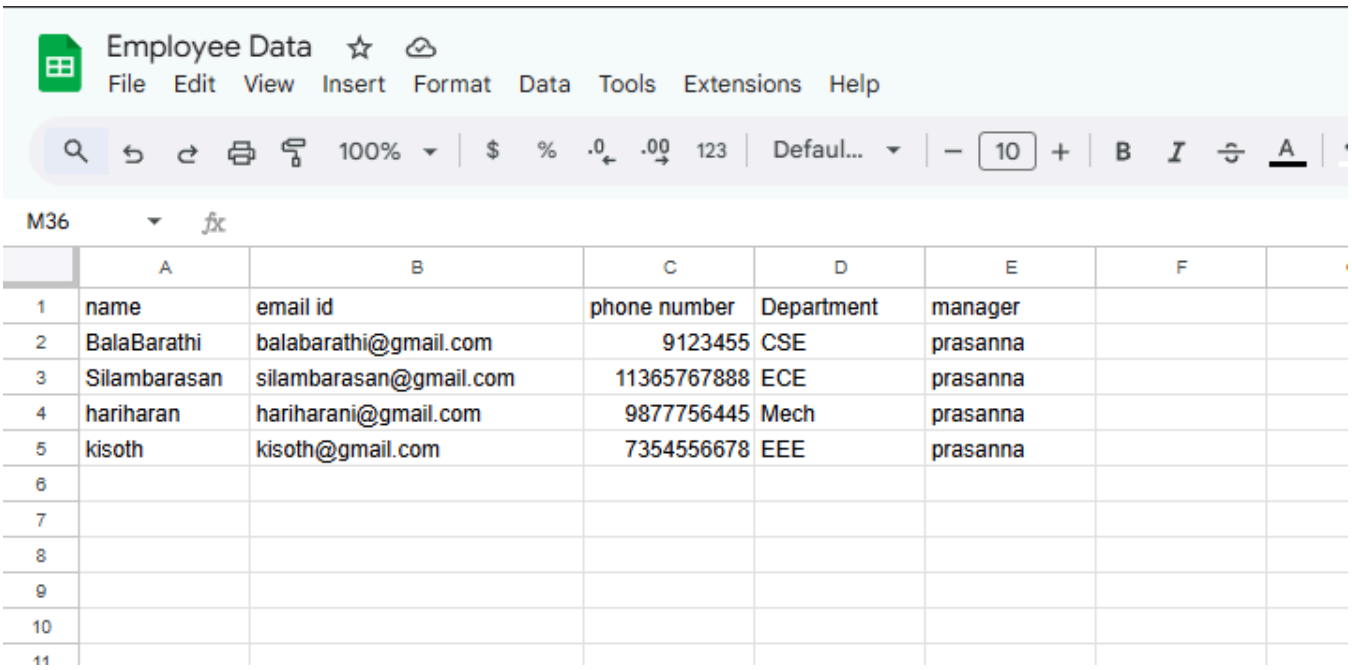
Activity 1: Create Records in Google Sheets

The first step is to prepare the employee data in a structured spreadsheet format.

1. Open Google Sheets (sheets.google.com).
2. Create a new spreadsheet and name it: employee data.
3. Add the following column headers in row 1: name, email id, phone number, Department, manager.
4. Enter at least 4 employee records with the sample data shown below.

name	email id	phone number	Department	manager
BalaBarathi	balabarathi@gmail.com	9123455	CSE	prasanna
Silambarasan	silambarasan@gmail.com	11365767888	ECE	prasanna
hariharan	hariharani@gmail.com	9877756445	Mech	prasanna
kisoth	kisoth@gmail.com	7354556678	EEE	prasanna

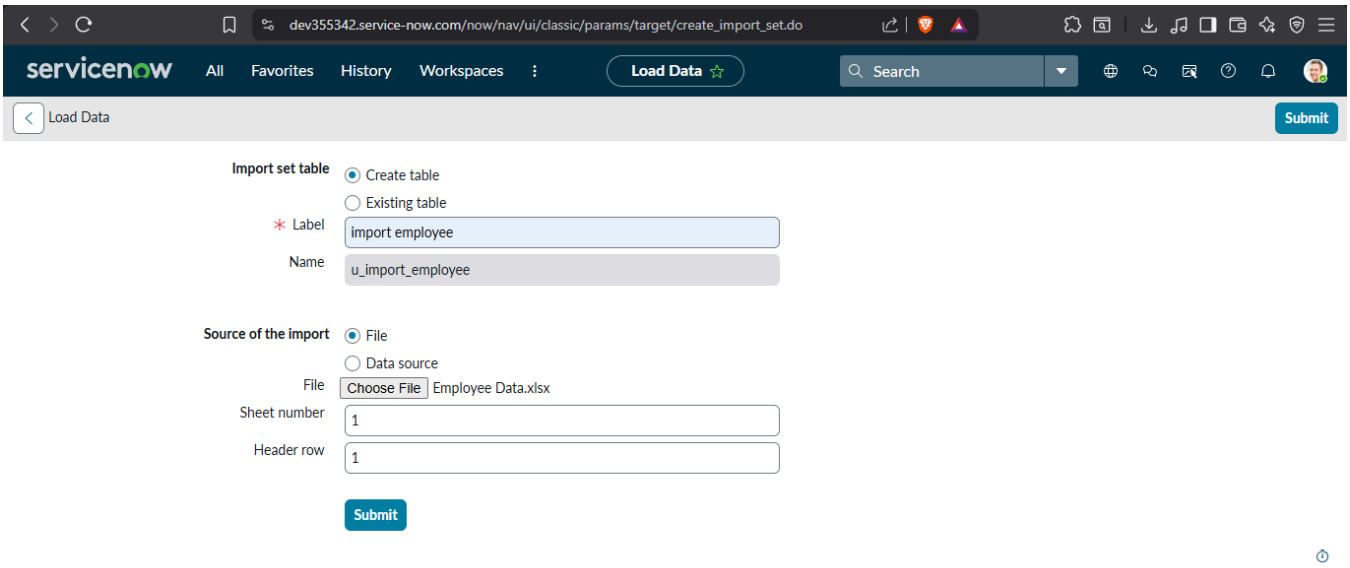
5. Go to File > Download > Microsoft Excel (.xlsx) or Comma Separated Values (.csv).
6. Save the file locally as emp.xlsx or emp.csv.



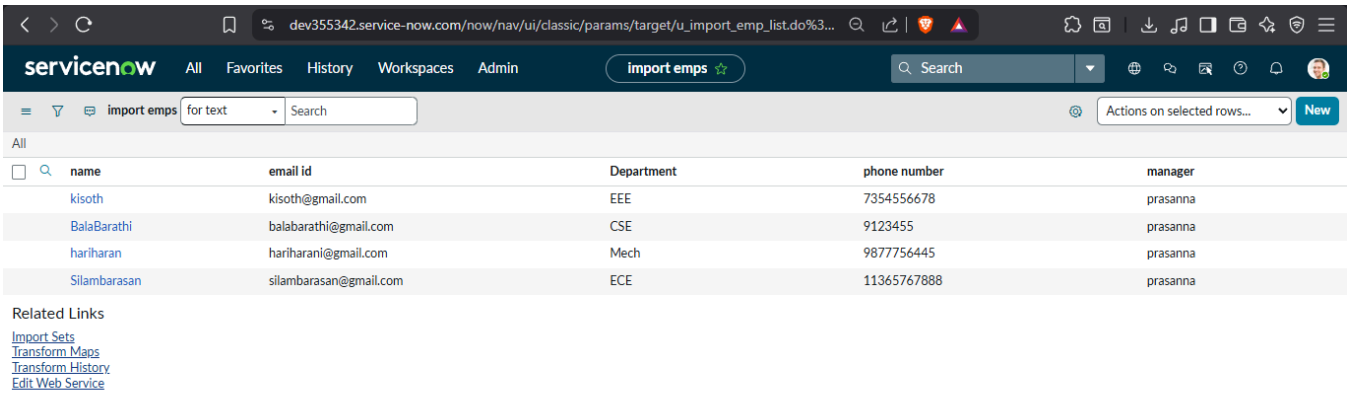
Activity 2: Import Data into ServiceNow

This activity loads the data file into a ServiceNow Import Set staging table.

7. Log in to your ServiceNow PDI instance.
8. Click on All in the top navigation bar.
9. In the search bar, type: import sets.
10. Under System Import Sets, click on Load Data.
11. On the Load Data form, select Create table under Import set table.
12. In the Label field, enter: import employee.
13. Under Source of the import, select: File.
14. Click Choose File and select the employee Excel/CSV file.
15. Set Sheet number to 1 and Header row to 1.
16. Click Submit to load the data.
17. Once loaded, click on the personalized list icon and add all fields to the view.



[Screenshot: Load Data form with file selection]



[Screenshot: Staging table showing imported records]

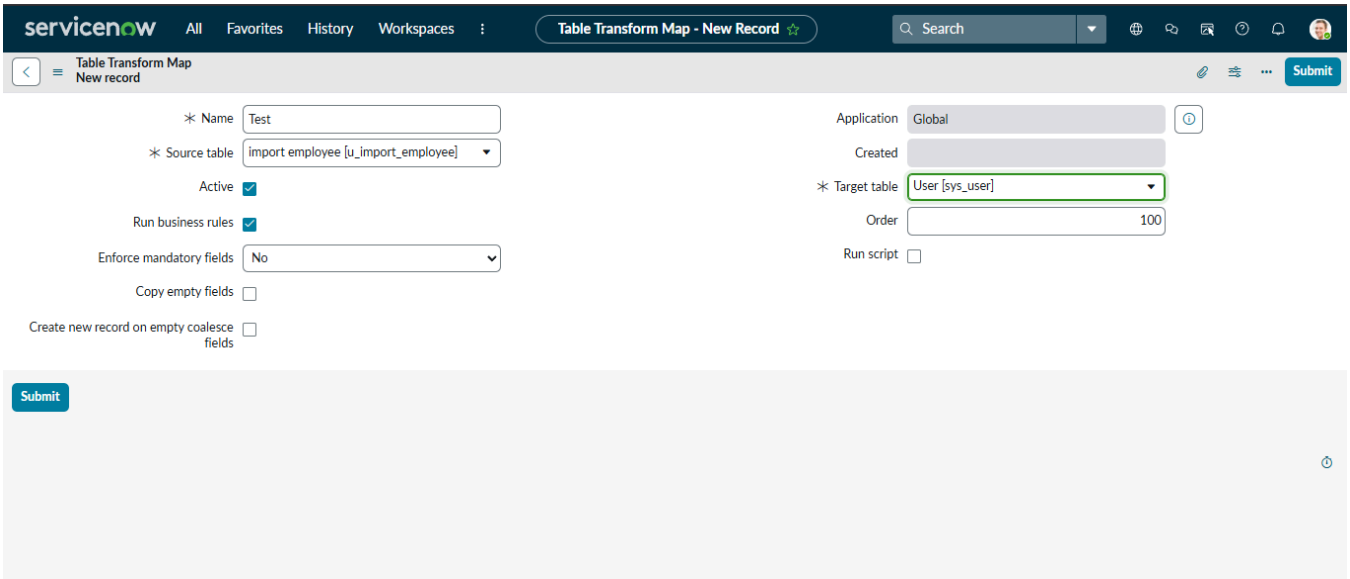
Activity 3: Create a Transform Map

A Transform Map defines how fields in the staging table map to fields in the target table.

18. In ServiceNow, click All and search for: import set.
19. Under System Import Sets, click on Transform Maps > New.
20. Fill in the following details:

Name	Test
Source Table	Import employee [u_123] (the staging table created in Activity 2)
Target Table	User [sys_user]
Active	Checked (Yes)
Run Business Rules	Checked (Yes)
Enforce Mandatory Fields	No

21. Click Submit to save the Transform Map.

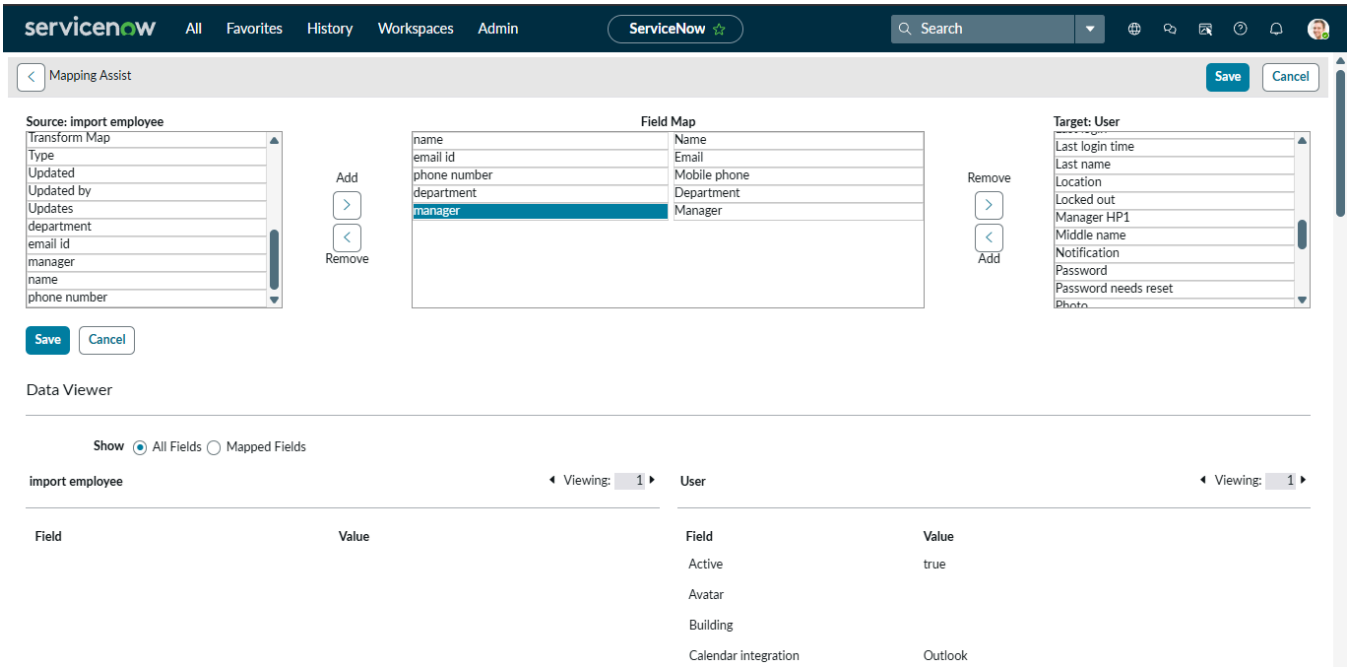


[Screenshot: Transform Map form filled in]

Activity 4: Match Fields Using Mapping Assist

This activity maps the staging table fields to the corresponding fields in the sys_user table.

22. Open the Transform Map created in Activity 3.
23. Scroll down to the Related Links section.
24. Click on Auto Map Matching Fields. ServiceNow will automatically map matching fields.
25. Click on Mapping Assist to manually map remaining fields.
26. In the Mapping Assist interface:
    - On the left panel (Source), locate the manager field and click Add.
    - On the right panel (Target), locate the Manager field and click Add.
27. Click Save to save all field mappings.
28. Verify the Data Viewer at the bottom shows the correct field mappings.

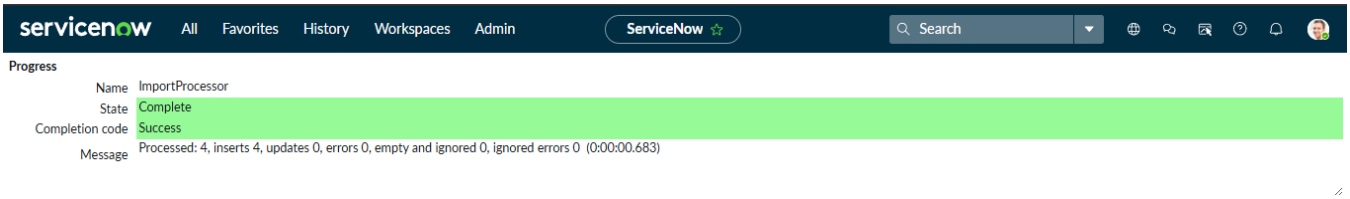


[Screenshot: Mapping Assist interface showing field mappings]

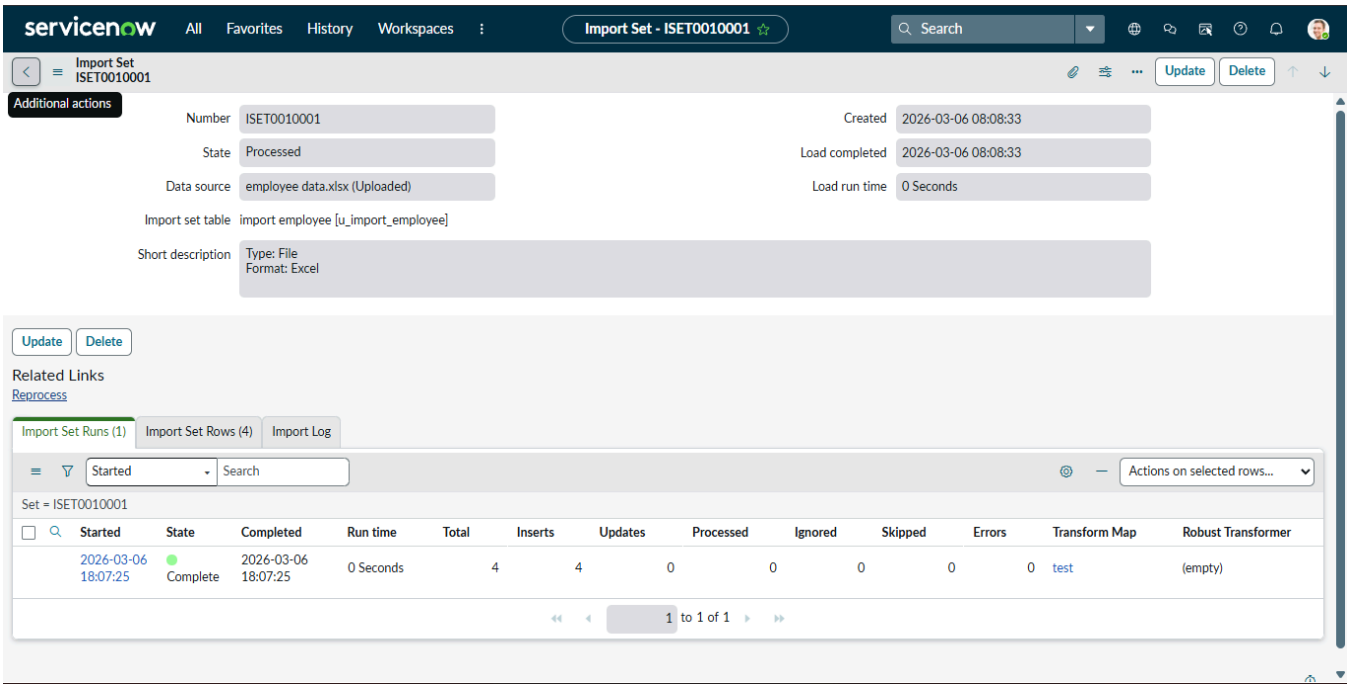
Activity 5: Transform the Data

This activity executes the transform operation to move data from the staging table to sys_user.

29. Open the Import Set record.
30. In the Related Links section, click on Transform.
31. Select the Transform Map 'Test - sys_user' and click the Transform button.
32. The progress screen will show: State: Complete, Completion code: Success.
33. Click on the Import Set number under Next Steps.
34. Click on the Import Set Rows tab to verify that 4 rows have been inserted with state: Inserted.



[Screenshot: Transform progress screen showing Success]



[Screenshot: Import Set Rows showing 4 rows Inserted]

Milestone 2: Dot Walking

Activity 6: Configure Dot Walking on the Incident Form

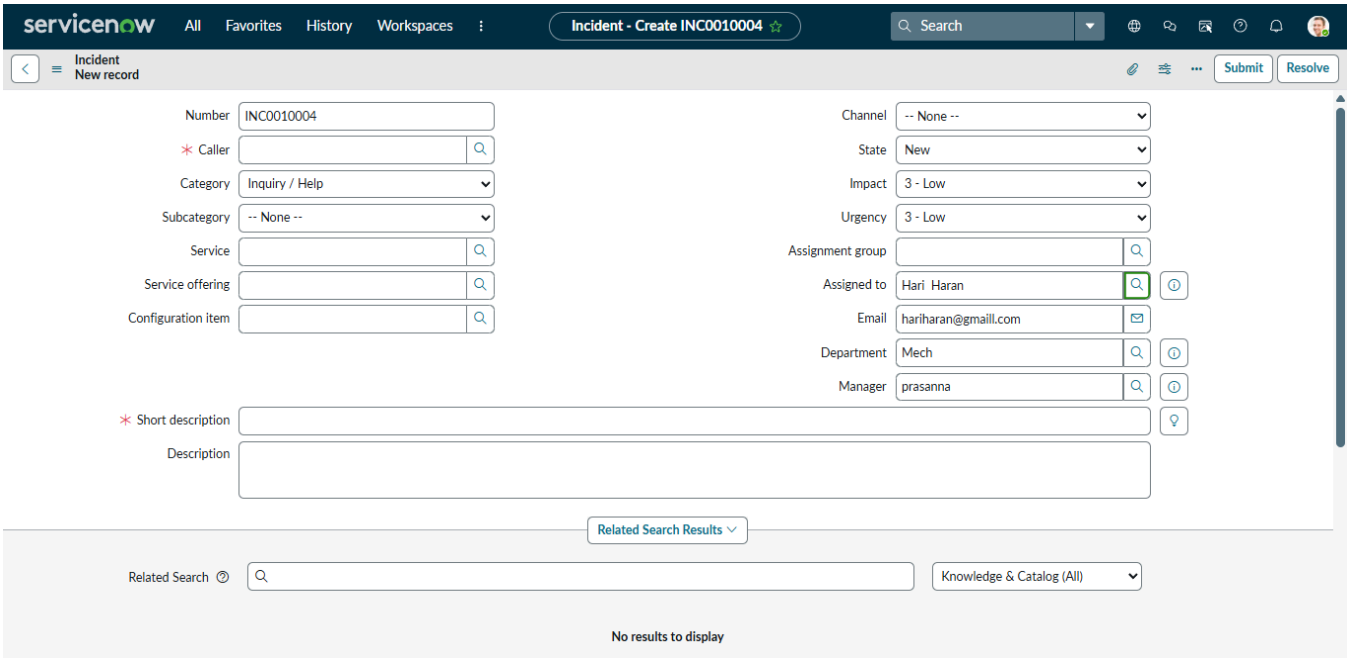
This activity adds dot-walked fields to the Incident form to automatically display employee relationship data.

35. In ServiceNow, navigate to Incident > Create New.
36. Observe that Department, Email, and Manager fields are not visible by default.
37. Right-click on the form header and click Configure > Form Layout.
38. In the Form Layout designer, find and select Assigned to Department and click > to add it to Selected.
39. Click Save.
40. Repeat to also add:

- Assigned to Email (assigned_to.email)
- Assigned to Manager (assigned_to.manager)

41. Click Save after adding all three fields.
42. Open a new Incident form.
43. Fill in mandatory fields (Caller, Category, Short Description).
44. In the Assigned To field, select one of the imported users (e.g., hariharan).
45. Verify that Department (Mech), Email (hariharan@gmail.com), and Manager (prasanna) are automatically populated.

[Screenshot: Form Layout designer showing dot-walked fields in Selected list]



[Screenshot: Incident form with auto-populated Department, Email, Manager]

2. Expected Results

Activity	Expected Outcome	Verification Method
Data Preparation	CSV/XLSX file with 4 employee records	File download successful
Import Data	Staging table with 4 rows populated	Import Set Rows tab
Transform Map	Fields correctly mapped to sys_user	Mapping Assist / Data Viewer
Transform Data	4 records in sys_user (State: Inserted)	Import Set Rows tab
Dot Walking	Auto-populated fields on Incident form	Create new Incident, assign user